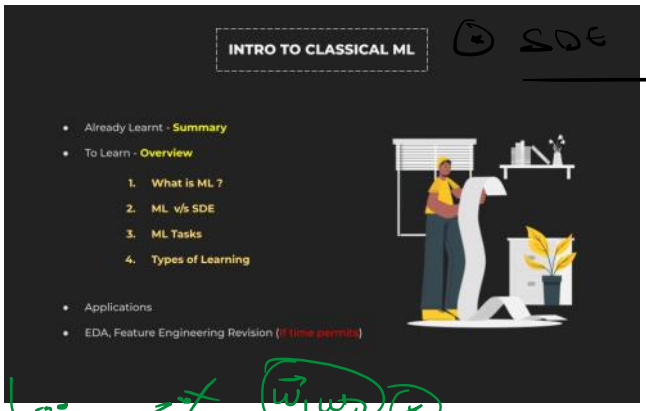
INTRO TO ML

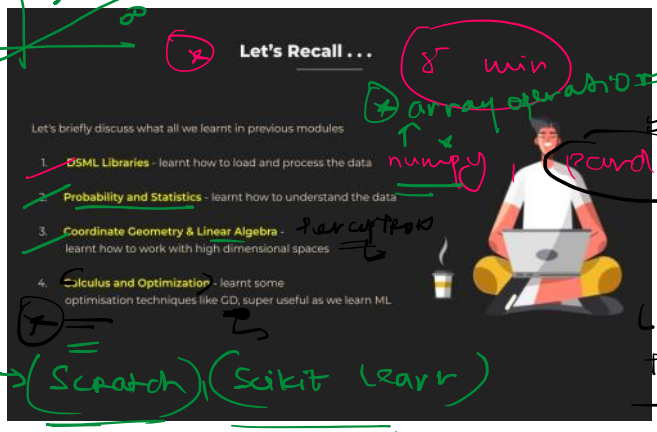
- Agenda →
  - DSML Libraries
  - Prob. Stats



SDE vs MLE

Categories Based on Tasks

- Classification
- Regression
- Rec System
- Reinforcement

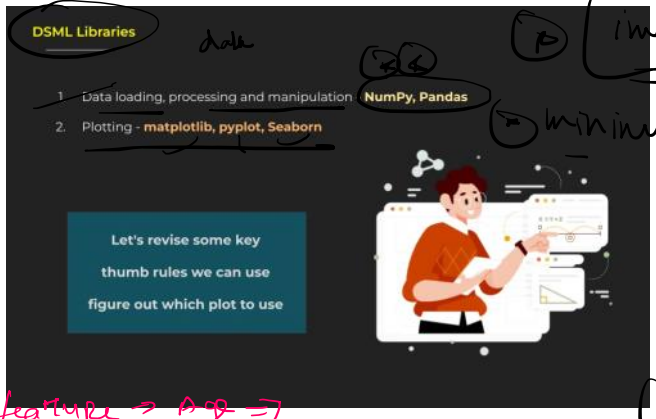


numpy, pandas, matplotlib, seaborn

Loads & Transform

alias

import pandas as pd

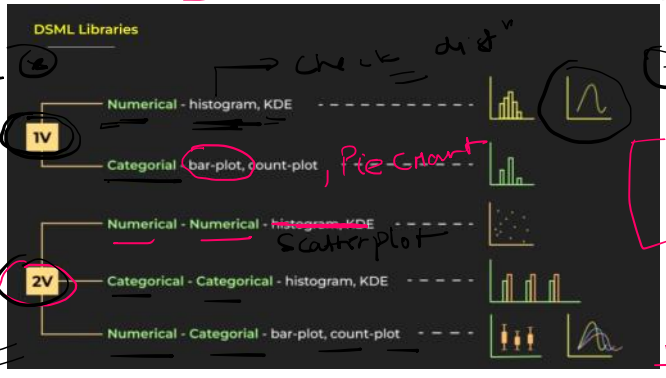


minimize or maximize

EDA

ML model

plot ? →



Predictions

model accuracy

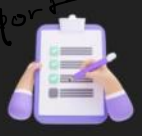
→ ERROR wt.

2V — Categorical - Categorical - histogram, KDE

Numerical - Categorial - bar-plot, count-plot

Department → HR, Tech, Mkt.

Plotting Examples :



Task

Patient(X) - Is diabetic ??

w h

features / Variables

Notice : Both the features are numerical

Which plot can we use here ?

- scatter-plot

Is there a way we can represent the third feature is_diabetic on this plot?

- We can perhaps color code the data-points - remember "hue" from Seaborn ?

Diabetic or Not

Hue →

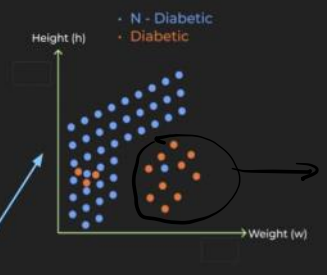
Diabetic

→ 3 variables

Technically, we have plotted three variables on this 2D plot

- numerical - numerical - categorical

3V



Diabetic

ML Prob stats where?

Numerical, Categorical

Probability & Statistics :

1. Remember we learnt how can we calculate things like P(diabetic)
   - Given some data, we can easily calculate this
   - Say 30% of the population is diabetic P(diabetic) = 0.3



2. We also learnt the concept of Conditional Probability
   - We can find how probability will change knowing that person has high weight,

   say w=120

   P(diabetic | w > 120) = 0.6

   Conditional Probability

Betting | Estimation

Fraud Detections

30% patients → Diabetic

Naive Bayes

Probability & Statistics :

Will it remain same if you get to know that the person with high weight is extremely tall?

- You would calculate **P(diabetic | h > 6, w > 120) = 0.25**



Data perspective . .

For ith patient

- Consider:

  height as first feature, **f1 = xi1**

  weight as the second feature, **f2 = xi2**

Height → 6ft.

Weight → 88kg

P(Diabetic | patient) = 77%

7ft. 88kg



- Consider:
  height as first feature, f1 = xi1
  weight as the second feature, f2 = xi2
- Predict if the patient is diabetic, yi
- We can predict, P(yi) instead

Concepts of P&S helps us in understanding that we can bring a probabilistic angle to this problem

Probability & Statistics :

In ML, we can think it like this

- Given feature of ith patient, xi=[xi1, xi1 , .. , xid]
- You have to predict P(yi)

Q. But how is it different from what we have been doing?
Q. We have calculated probability before also, right?



Case 1:
xi is present in the data --> becomes a Data Analysis problem, simply calculate the P(y) using data

Case 2:
xi is not present in the data --> becomes a ML problem --> "predict" P(y) using an ML model

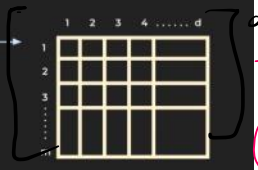
Linear Algebra :

You have some data X with features - height, weight, age, gender...x_{d}

- You can represent it data for one sample using a vector

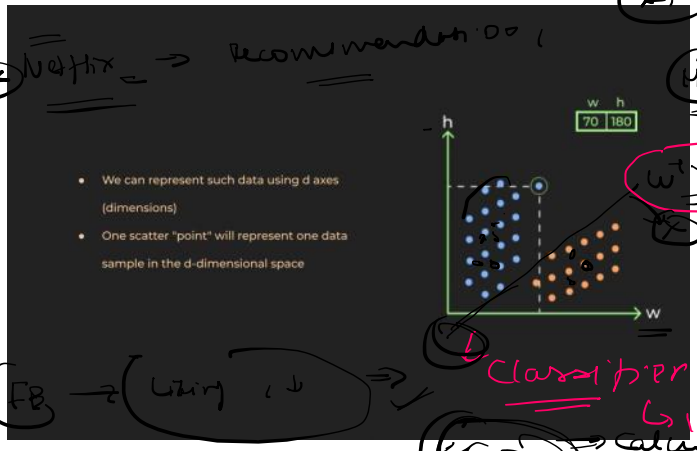


- And data for m samples using a 2D - Matrix



In data language, this is d-dimensional data, not a 2D data

1. In **Maths** - we call this vector as d-dimensional vector
2. In **Numpy** - we call it as a 1D array or a vector with d-elements



- We can represent such data using d axes (dimensions)
- One scatter "point" will represent one data sample in the d-dimensional space

POINTS TO REMEMBER



- We use numpy and pandas for data loading, processing and manipulation
- The type of plot can be decided looking at the kind of data we have.
- In conditional probability, we check the probability of an event with respect to another event which has happened.
- If xi is present in the data, it becomes a data analysis problem, else it becomes a ML problem.

happened.

- If xi is present in the data, it becomes a data analysis problem, else it becomes a ML problem.



ML v/s Classical Programming:

What is ML ?

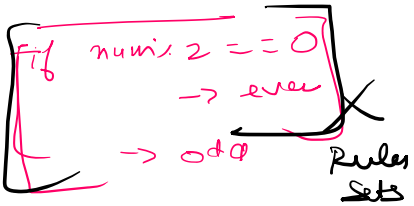
Definition by Arthur Samuel -

ML provides ability to learn without being explicitly programmed

Q.1 - But can a "dumb" computer perform anything by itself if don't tell what to do?

Q.2 - If somehow, let's assume that it is possible, shouldn't we use ML for everything?

Think : Why do we have SDEs then? Why not all of them MLEs?



ML say → generic algo

i/p → Test data → o/p

learning

SDE pipeline

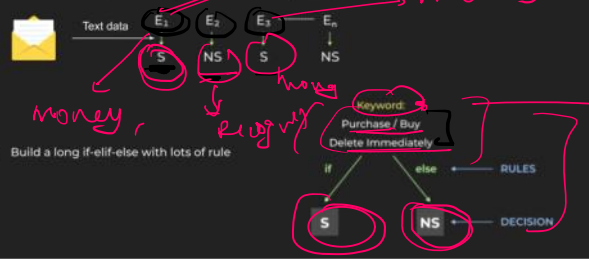
Lets understand with a task

Given an email, you have to identify whether is spam or not spam

Example: Email → Spam vs Not Spam

How does the SDE pipeline look ?

1. Look at the data, find some "patterns" - certain keywords, certain phrases



2. Build a long if-elif-else with lots of rule

money, money, money

flaws ?

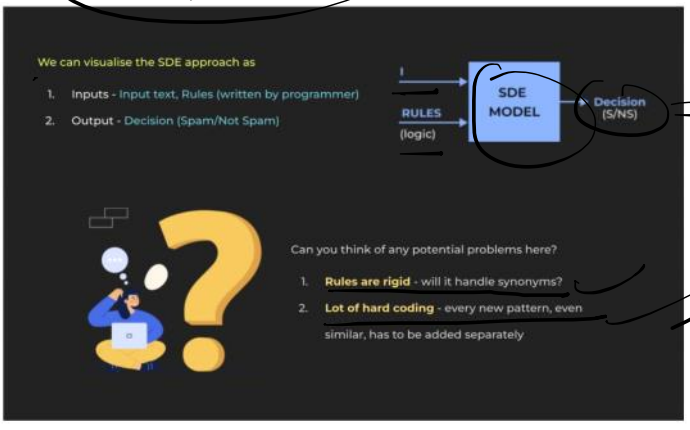
SDE pipeline

Rule set

1000s | 1000s

[ ]

ML pipeline

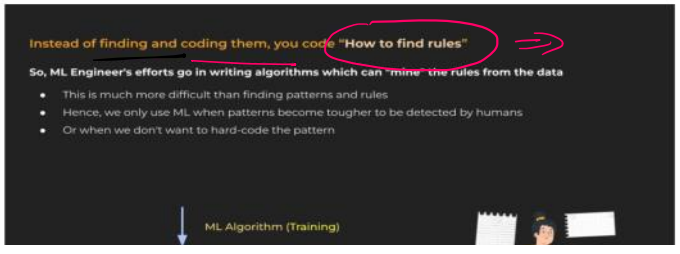


→ S | NS

not adaptive

NS → Spam

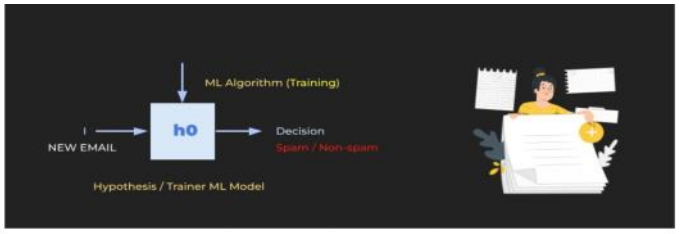
NS → Gmail

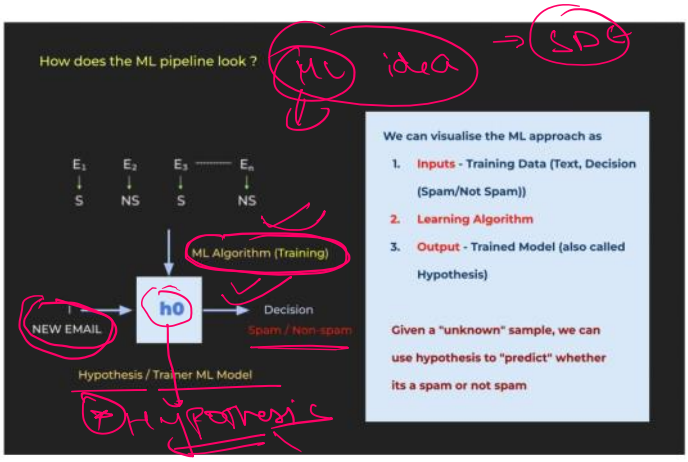


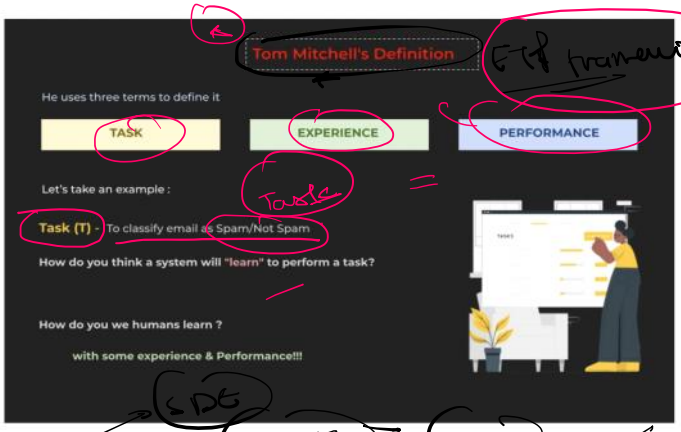
1 Million emails

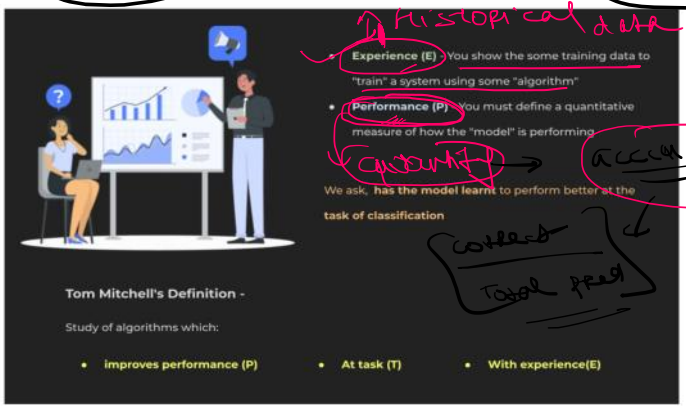
E1 → S

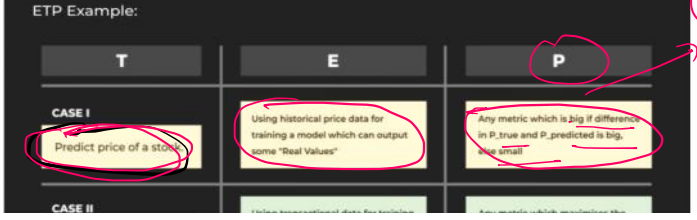
E2 → NS











E_1
E_2 → NS
E_3 → S
:
E_{IN} → NS

multiple approaches

1st → money → Spam

2nd → money → credit → NS

3rd

multiple patterns

SDE V/S MU

Algorithm →

maths

Best

ID: 17

Tpain → i/p → Email

→ label → S/NS

ML algo → Log Reg

↓ Hypothesis

↳ Trained model

i/p → ML algo → o/p

h0 → trained model

profit →

Sensors data → DE

Json → csv /

Predict price of a stock	Using historical price data for training a model which can output some "Real Values"	Any metric which is big if difference in P_true and P_predicted is big, else small
CASE II Segment Customers	Using transactional data for training a model which can put similar customers into same cluster	Any metric which maximises the inter cluster difference and minimises the intra cluster distance
CASE III Categorise Image into orange, kinno, mossambi	Train using Images + Labels using a model which can "classify" an image as O, K, M	Any metric which gives a high(er) score for correct label, and low(er) score for incorrect labels

POINTS TO REMEMBER

- SDE pipeline : Provide Rules and Input, Model will give decision.
- ML Pipeline : We provide Input data and ML algorithm, algorithm will mine rules itself and decide output based on them
- Tom-Mitchell def : Study of algorithms which improves performance (P) at Task (T) with Experience (E)
- Types of task based on task - classification, regression, clustering, recommendation, forecasting



Based on the type of "task" ML Algos can be categorised as

Classification: classify into one of categories, e.g: citrus 🍊 fruits classification

Regression : predict a real value, e.g, price prediction using stock fundamentals

Clustering: Group similar samples, e.g: customer segmentation

Recommendation: A different task e.g: recommend a movie🎥

Forecasting Forecasting stock prices based on last 10 days prices

Based on the type of "learning" use in training the model

Supervised Learning: Output labels used for training

Unsupervised Learning: Output labels are not used

Reinforcement Learning: To be discussed later



CLASSIFICATION

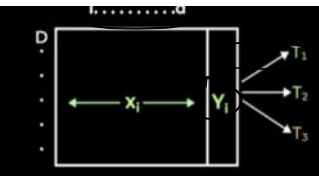
Recalling the fish 🐟 Sorting example

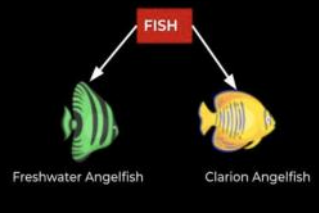


Recalling the fish 🐟 Sorting example

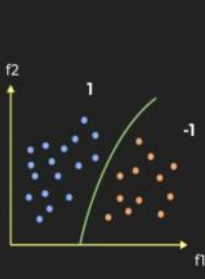
1. X being d-dimension features, with m samples
2. Y is categorical variable: T1,T2...Tn
   - if Y only have T1,T2 - Binary
   - if Y T1,T2,...,Tn - Multiclass
   - this shows,

Classification has its own subtypes





- In the fish🐟 sorting example,
  - if only 2 types of fish to sort: Binary
  - If 3 or 4 or n number of fish: Multi-class
- Now Classification is deciding a hyperplane that separates the points of the data correctly





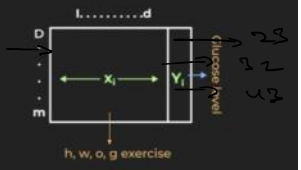
Y_i = {-1, 1} - categorized

$$D = \{(x_i, y_i)_{i=1}; X_i \in R^d, y_i \in \{0,1\}\}$$

Target is categorical

REGRESSION

- If we had **d-dimension** features of blood samples of m patients
- Now instead of Y being **Diabetic/ non-diabetic**, Y is blood glucose level this becomes a **Regression problem**
- What is the **Range (y)**?
  - **Numeric** (-inf, inf)
  - **Real Numbers**
- Now the task will be, we want a hyperplane that is close to the data points for us to predict **Blood Glucose level for new patients.**



$$D = \{(x_i, y_i)_{i=1}; X_i \in R^d, y_i \in R\}$$

Target is real valued number.

Regression V/S Classification



The main **DIFFERENCE** between Regression and Classification is:

- Y is **Real numbered** in Regression while Y is **categorical** in Classification
- In Regression **we want hyperplane close to the data points for prediction while in**
- Classification, **we want a hyperplane that separates points of two or more classes accurately**

Regression V/S Classification



2. The thing **COMMON** between them is

COMMON: Yi is present in data.

$$D = \{X_i, Y_i\}^m - Labelled\ Data$$

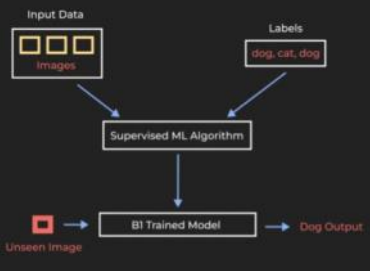
Supervised learning - uses "labelled data" for training the ML model

POINTS TO REMEMBER

- Types of task based on learning - supervised, unsupervised, reinforcement learning
- Output in regression is continuous (numeric) while in classification is categorical
- Both regression and classification are supervised since labelled data is present

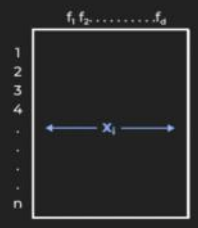


Supervised Learning



- Supervised models learn from the labeled dataset
- For the training procedure, the input is a known training data set with its corresponding labels
- The learning algorithm produces a function/hypothesis to finally make predictions about some new unseen observations.

Clustering



- Imagine 100M Amazon customers with features like location, average $ dollar value per purchase



Whether a customer is a prime member or not ?

- **There are no labels** yi here - we are looking to find the hidden labels of these data-points
- Since there are no yi involved in the training data D, this is an example of **Unsupervised Learning**



This is called **Clustering** - **means to group the similar data-points (customers)**

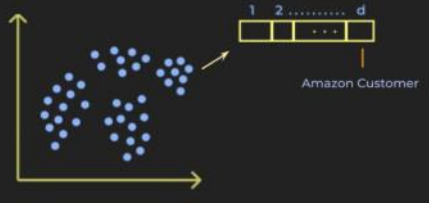


Why would we want to cluster customers?

1. Imagine we have to carry-out qualitative survey - instead of surveying all, we can survey one representative customer
2. We can create group specific strategy - offer more discount to certain group

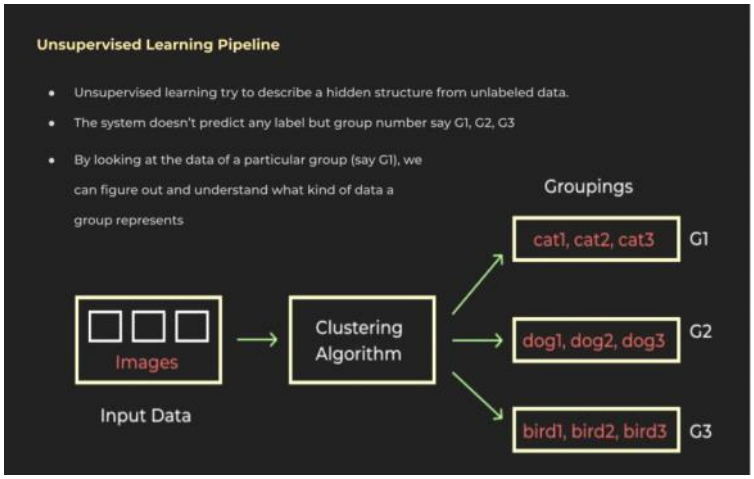
Unsupervised Learning

Notice that the customer data has been grouped into clusters - each data point can be represented in d-dimensional space



Q. How can we cluster similar points?

- Similar points have less distance between them - we can use some relevant distance metric to calculate these distances



Question

Which of the following provides the best explanation of Machine Learning?

Choices

- [ ] Machine learning learns from labelled data
- [ ] Machine learning is the field of giving robots the ability to act intelligently.
- [x] ML enables computers to learn without explicit programming
- [ ] Machine learning is the science of programming computers.

unlabelled data

Dog → walk → Treat / punishment → RL

Reinforcement learning

Task → correctly → reward

Agent

feat type, mileage → CNG, petrol, Diesel → x_i → L → Penalty

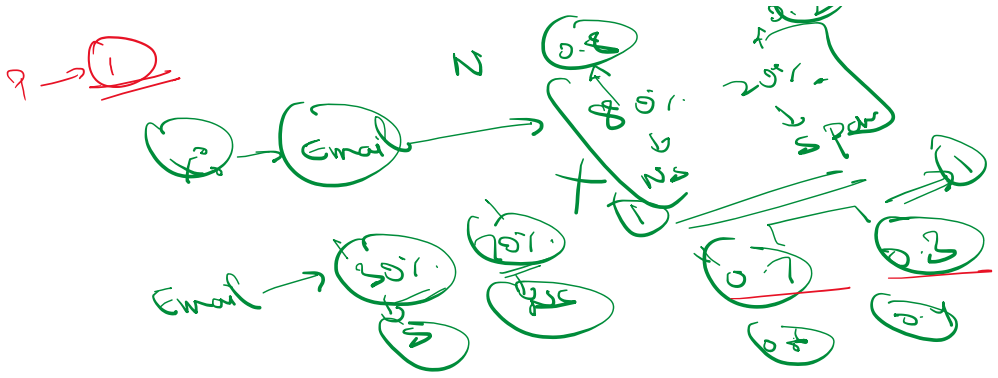
Question

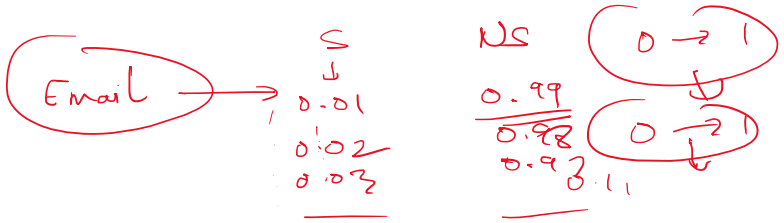
Which of the followings is/are true about classification?

Choices

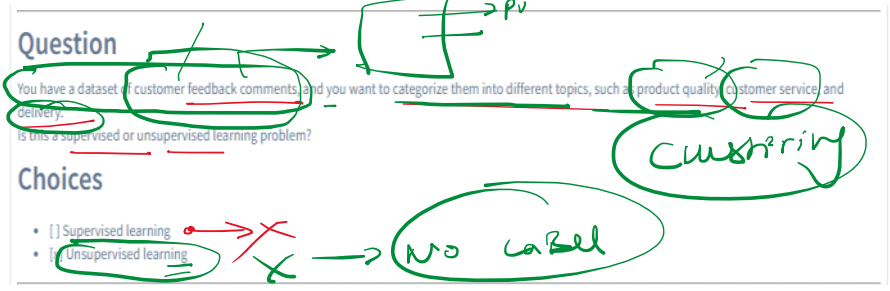
- [ ] Classification can be defined as a predictive model mapping inputs to discrete outputs
- [ ] Class label prob. enables classification algos to predict continuous values.
- [ ] A classification algorithm can have both discrete and real-valued input variables.
- [x] All of the options

D_i, R_i → B / NB / Logistic / Classification

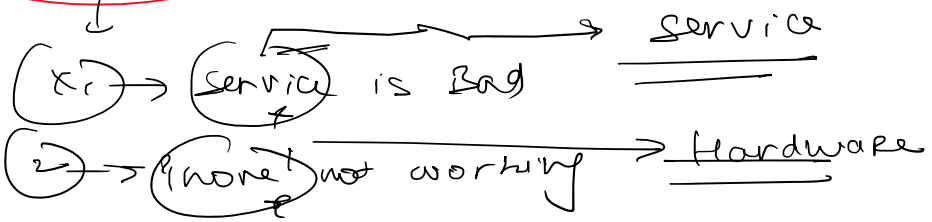


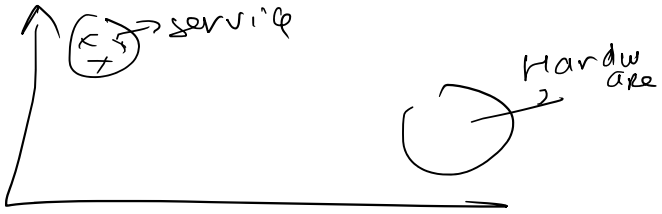


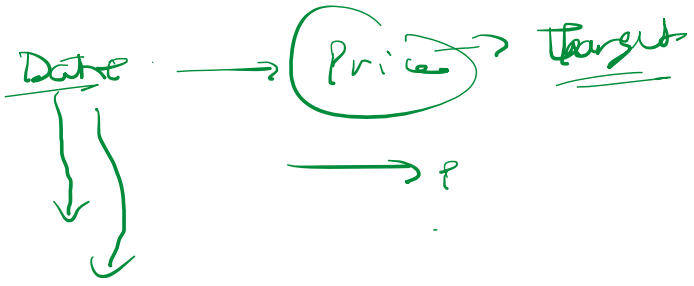
keyword Based











Question

You are working on a project to predict stock prices based on historical market data, and you have a dataset with features such as past stock prices, trading volume, and other market indicators.
Is this a supervised or unsupervised learning problem?

Choices

- [x] Supervised learning
- [ ] Unsupervised learning

title: Quiz 8
description:
duration: 30

